

Website Traffic Analysis

Prepared for Alfido Tech | Task 3 of 3 — Data Analytics Internship Program

Executive Summary

This report analyzes 226,230 website traffic events (pageviews, previews, and clicks) recorded across music smart-link pages over one week (Aug 19–25, 2021), covering 211 countries, 2,419 artists, and 3,822 individual links. Total traffic declined 13.9% across the week, with pageviews and clicks falling in lockstep. The overall click-through rate is a strong 39.2% of pageviews, though only 20.1% of visitors preview content before converting — most traffic appears referral-driven rather than discovery-driven. Traffic is geographically concentrated in Saudi Arabia and India (~40% combined), and heavily dependent on a single track, which alone drives 18% of all traffic. Click-through quality varies widely by country and doesn't track with volume — the highest-volume market (Saudi Arabia) converts below average, while several smaller markets convert far better. Five actionable recommendations are provided in Section 6.

Total Events	Pageviews	Click-Through Rate	Countries	Weekly Change
226,230	141,980	39.2%	211	–13.9%

Key Findings

- Total traffic declined 13.9% across the one-week window, with pageviews and clicks falling in lockstep while previews stayed flat.
- Overall click-through rate is a strong 39.2% of pageviews, but only 20.1% of visitors preview first — traffic appears largely referral/share-driven rather than discovery-driven.
- Traffic is geographically concentrated — Saudi Arabia and India together account for ~40% of all traffic; within Saudi Arabia, two cities (Jeddah, Riyadh) dominate.
- Click-through quality doesn't track with volume — Saudi Arabia has the most traffic but a below-average 32.9% CTR, while smaller markets like Pakistan (51.5%) convert far better.
- Traffic is highly concentrated in a single piece of content — one track (“Jalebi Baby”) drives 18% of all traffic.

Top 5 Recommendations (see Section 6 for full detail)

1. Investigate the cause of the 13.9% weekly traffic decline.
2. Prioritize high-CTR, currently-smaller markets for growth investment.
3. Diagnose why Saudi Arabia's CTR lags its traffic volume.
4. Reduce dependency on a single top track via cross-promotion.
5. Study and replicate what the highest-CTR top links do differently.

1. Introduction & Data Overview

Dataset: Music smart-link traffic events — *traffic.csv*. Each row represents a single traffic event (a pageview, preview, or click) recorded on a music smart-link page, along with the visitor's country/city and the artist/track/album/link being viewed. There is no user or session ID, so each row is best understood as one atomic event rather than one visitor.

Column	Description
event	Event type: pageview, preview, or click
date	Date of the event (daily granularity, no timestamp)
country / city	Visitor's geography
artist / album / track	Content being viewed
isrc	International Standard Recording Code (not used in this analysis)
linkid	Unique identifier for the smart-link page

Scope: 226,230 clean events across 211 countries, 11,992 cities, 2,419 artists, 3,547 tracks, and 3,822 links, spanning exactly one week: 2021-08-19 to 2021-08-25.

2. Data Cleaning & Methodology

- **Duplicate rows (~46% of the dataset):** verified to be real repeat traffic, not a data error. With no user ID, a popular link viewed by hundreds of different anonymous visitors from the same city on the same day produces identical rows — confirmed directly (e.g. one track's link was viewed 729 times from Jeddah, Saudi Arabia on a single day). Duplicate rows were therefore **not removed**, since doing so would silently delete real traffic volume.
- **Missing values:** country/city/artist/track missing for a small number of rows (11–37 out of 226K) — dropped, since they can't be attributed to any analysis dimension. The isrc field is missing for ~3% of rows but isn't used in this analysis.
- **“Unknown” city:** ~16,661 rows have city = “Unknown” (geolocation blocked or undetermined) — a legitimate category, excluded only from the city-level ranking since it isn't an actual city.
- **Date parsing:** the data spans exactly seven days (2021-08-19 to 2021-08-25) — a single week, so this analysis reports day-level trends rather than monthly/seasonal patterns.

3. Traffic Trend Analysis

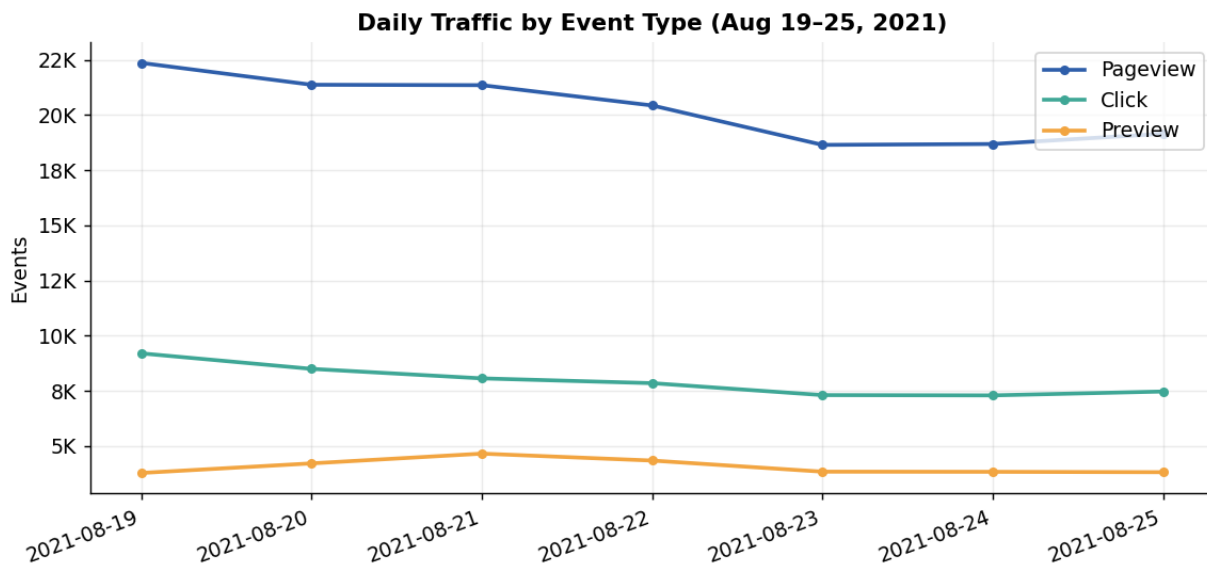


Figure 1. Daily traffic by event type, Aug 19–25, 2021. Pageviews and clicks decline in lockstep while previews stay comparatively flat, pointing to a broad decline rather than a shift in on-page behavior.

Total daily traffic fell from approximately 35,300 events on August 19th to about 30,400 events on August 25th — a 13.9% decline over the seven-day window. Since this dataset covers only one week, it isn't possible to say from this data alone whether this is a normal weekly pattern, a cooldown from a viral spike, or an external factor — this is flagged as the top item for further investigation in Section 6.

4. Conversion Funnel Analysis

A pageview is a visit to the smart-link page. From there, a visitor can preview (play a short audio clip) and/or click through to a streaming platform. These aren't strictly sequential for every visitor, but pageview is always the entry point.

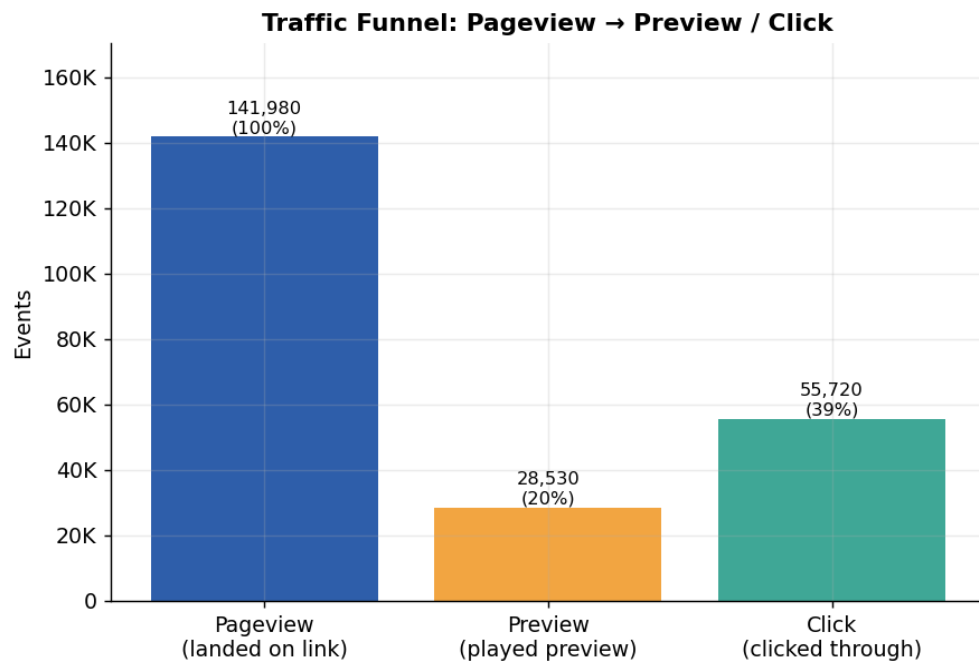


Figure 2. Traffic funnel from pageview to preview/click. 39.2% of pageviews convert directly to a click-through.

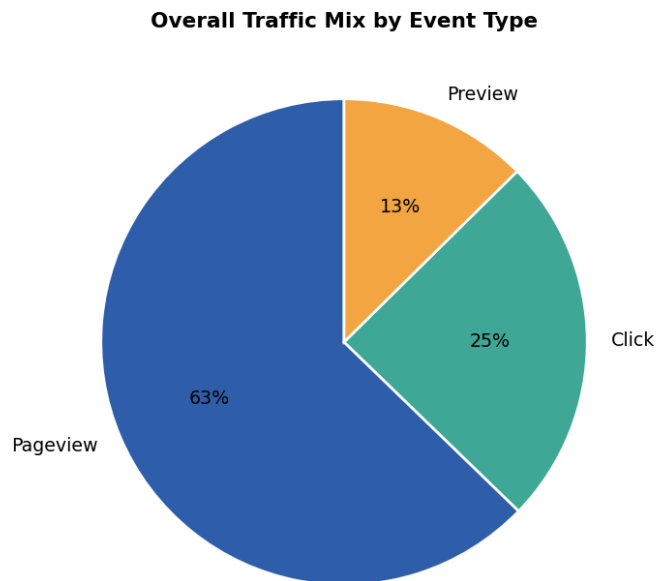


Figure 3. Overall traffic mix. Pageviews make up 63% of all events, clicks 25%, previews 13%.

5. Geographic Analysis

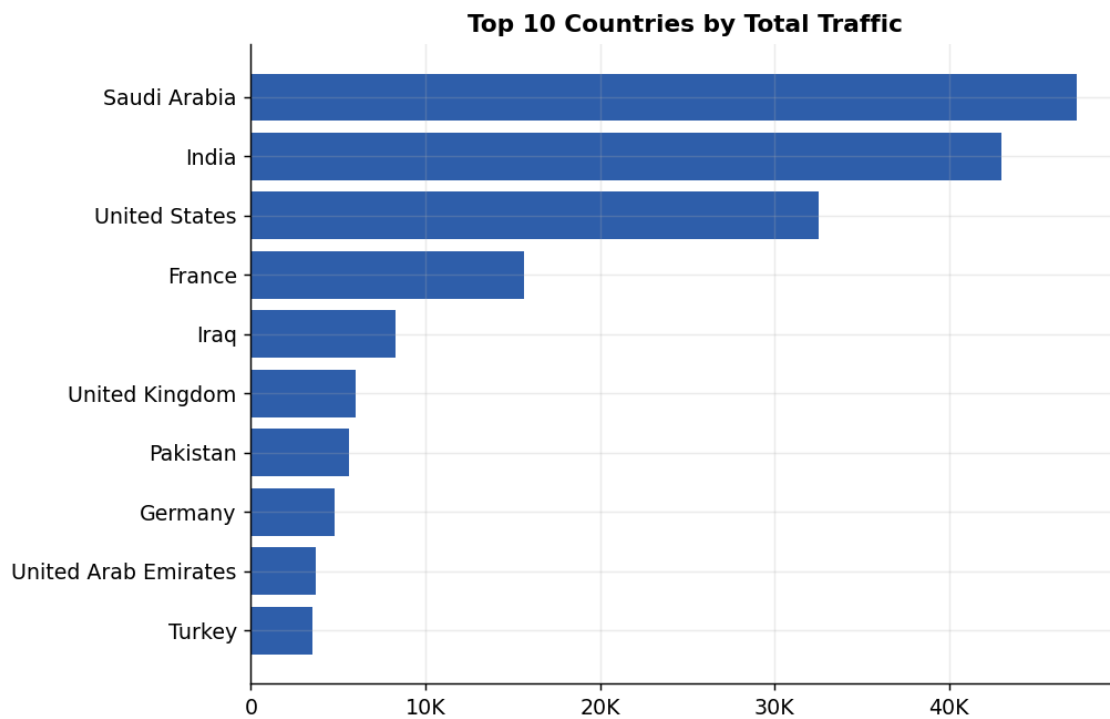


Figure 4. Top 10 countries by total traffic. Saudi Arabia and India lead by a wide margin.

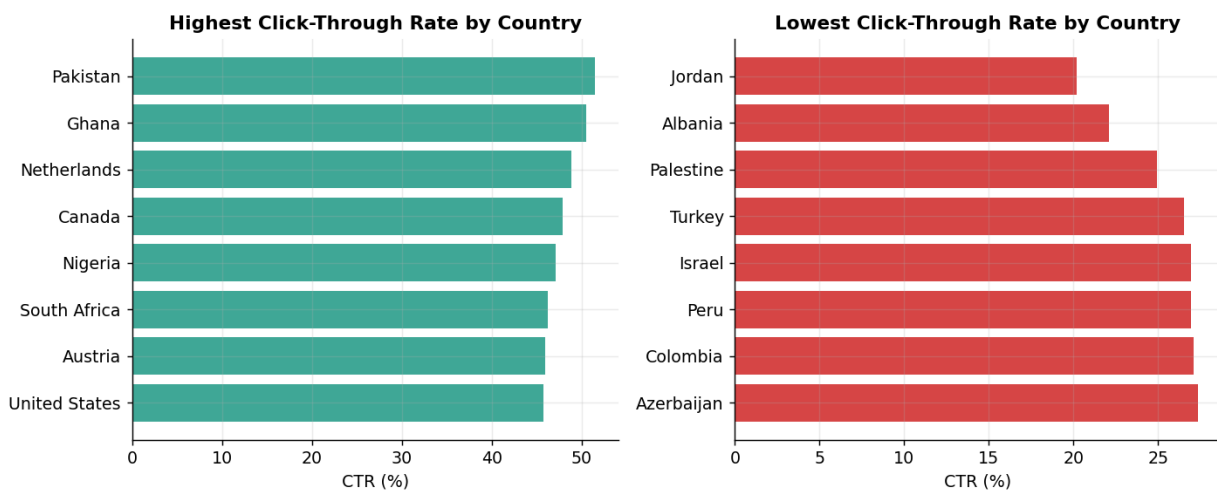


Figure 5. Highest and lowest click-through rate by country (min. 200 pageviews). CTR varies from 20% to 51% and does not track with traffic volume.

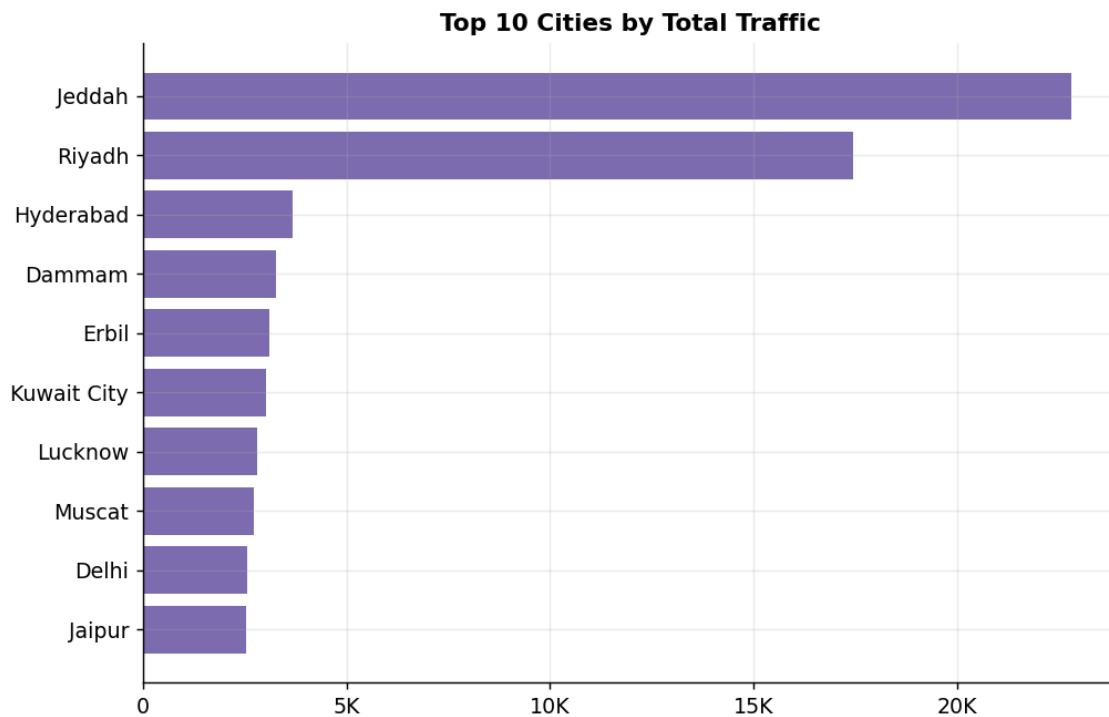


Figure 6. Top 10 cities by total traffic (excluding “Unknown”). Jeddah and Riyadh, both in Saudi Arabia, dominate.

6. Content Performance (Artists, Tracks, Links)

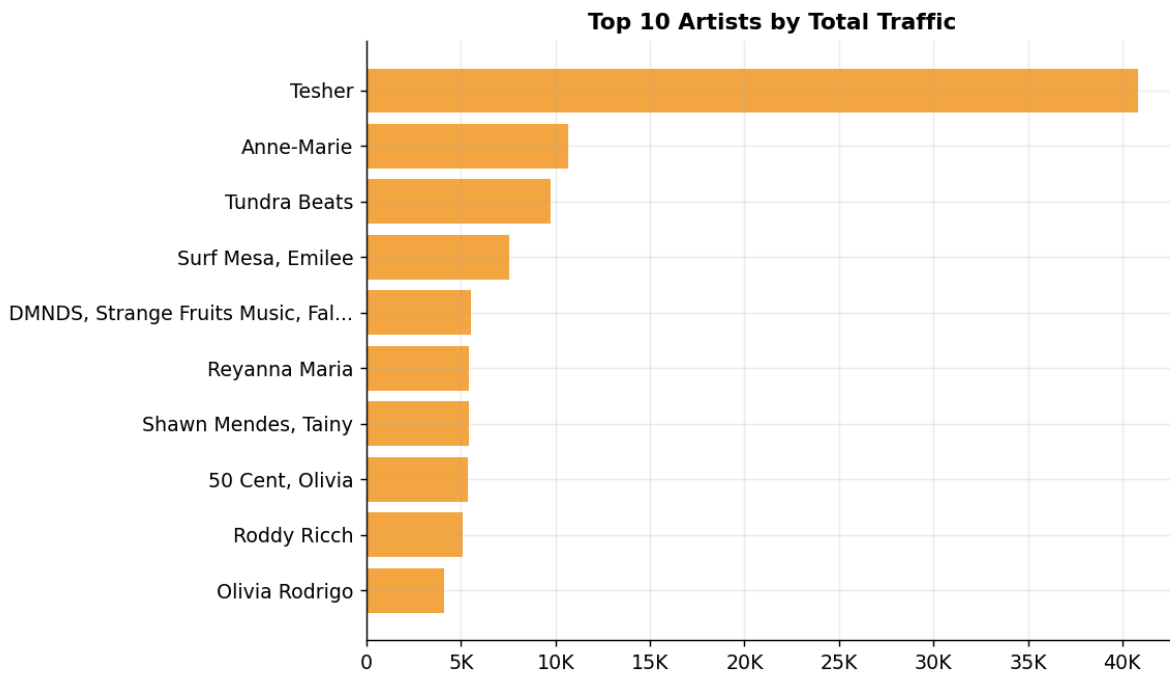


Figure 7. Top 10 artists by total traffic. Teshar's “Jalebi Baby” alone drives 40,841 events — 18% of all traffic in the dataset.

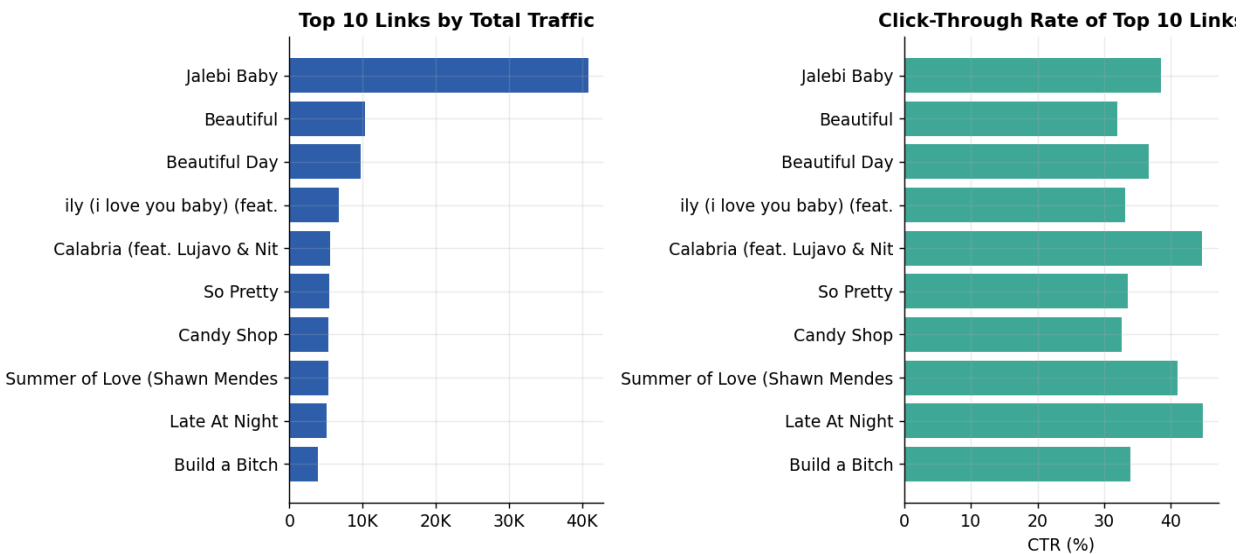


Figure 8. Top 10 links by traffic volume and their click-through rates. CTR ranges from 32.6% to 44.9% even among the top 10 — a meaningful spread that isn't explained by popularity alone.

7. Actionable Recommendations for Alfido Tech

1. Investigate the cause of the 13.9% weekly traffic decline

Determine whether this is the natural cooldown of a viral moment tied to the top track, a broader external factor (e.g. a promotional campaign ending), or a normal weekly pattern that a longer data window would clarify. Understanding the cause lets Alfido Tech plan around it rather than be surprised by future dips.

2. Prioritize high-CTR, currently-smaller markets for growth investment

Pakistan (51.5%), Ghana (50.5%), Netherlands (48.8%), and Canada (47.8%) all convert well above the 39.2% overall average. Growing traffic acquisition in these markets is likely more efficient than pushing further into already-saturated, lower-converting Saudi Arabia.

3. Diagnose why Saudi Arabia's CTR lags its traffic volume

Saudi Arabia drives the most total traffic (47,334 events) but converts at a below-average 32.9%. Even a modest CTR improvement there — through localized page design, language, or content placement — would yield an outsized number of additional clicks given the sheer volume involved.

4. Reduce dependency on a single top track

“Jalebi Baby” alone drives 18% of all traffic. Actively cross-promoting other artists and tracks on high-traffic pages (e.g. “you might also like” recommendations) would diversify the traffic base so a future decline in one song's popularity doesn't disproportionately affect total numbers.

5. Study and replicate what the highest-CTR top links do differently

Among the top 10 links by volume, CTR still ranges from 32.6% to 44.9%. Since these are all already popular tracks, this spread likely reflects differences in page layout, preview placement, or call-to-action design rather than the content itself — identifying and replicating the best-performing patterns could lift conversion broadly across the platform.

Prepared as part of the Alfido Tech Data Analytics Internship Program — Task 3: Website Traffic Analysis. Full code, methodology, and additional exploratory visualizations are available in the accompanying Jupyter notebook (Website_Traffic_Analysis.ipynb).